

Enterprise SOC Home Lab: Wazuh SIEM, Active Directory & Multi-OS Agent Operation

This report documents an operational enterprise-style SOC home lab built inside an isolated VMware network. The environment includes Active Directory, Windows endpoints, Linux endpoints, internal mail infrastructure, Wazuh SIEM/XDR, and curated evidence screenshots. Passwords, generated Wazuh credentials, private keys, and setup screens containing secrets are excluded.

Enterprise SOC Home Lab - Operational Multi-OS Monitoring

Active Directory, corporate mail, Linux integration, Wazuh SIEM/XDR, deployed agents

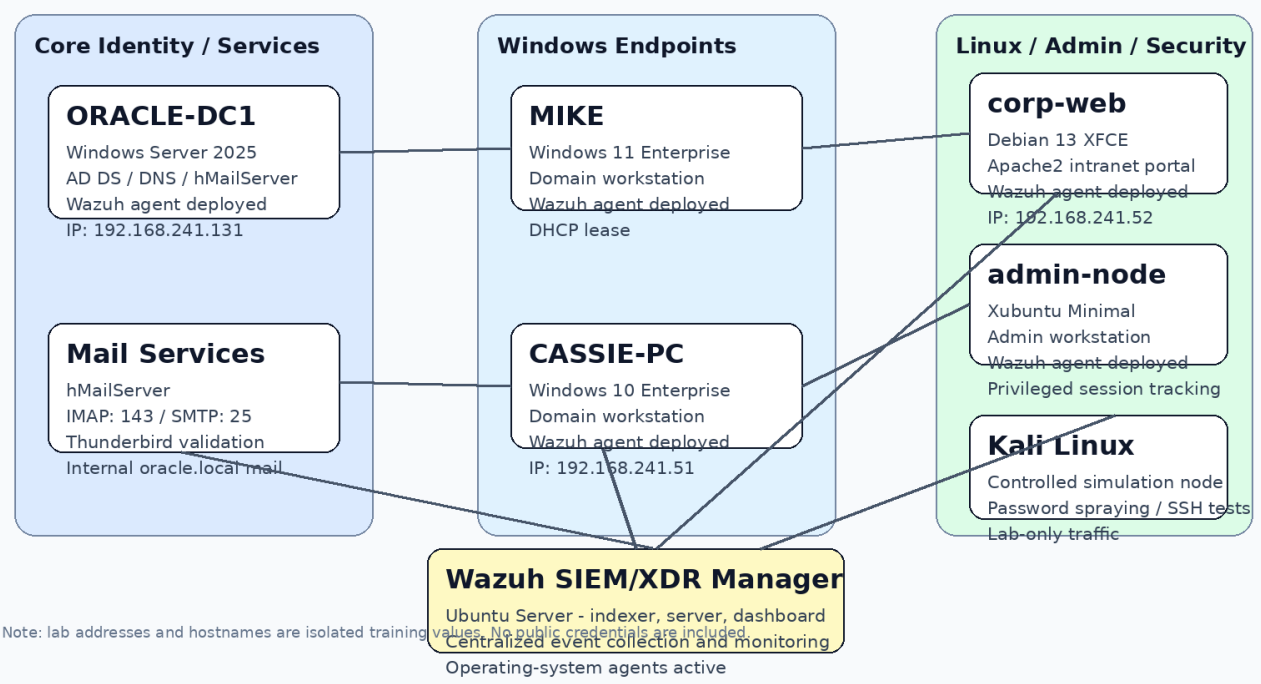


Figure 1. Updated topology: agents are deployed on the operating systems that generate endpoint telemetry.

1. Current Operational Status

Layer	Status
AD/DNS	ORACLE-DC1 deployed as domain controller and DNS for oracle.local
Windows agents	Windows Server, Windows 10, and Windows 11 endpoints monitored by Wazuh agents
Linux agents	Debian corp-web and Xubuntu admin-node monitored by Wazuh agents
Mail	hMailServer and Thunderbird validated for internal IMAP/SMTP flow
Wazuh	Manager/indexer/dashboard used for centralized telemetry and threat hunting
Simulation	Kali reserved for lab-only password spraying, brute-force, and privilege-escalation validation

2. Active Directory and Windows Endpoint Layer

ORACLE-DC1 provides Active Directory Domain Services, DNS, and the internal mail server role. Windows 10 and Windows 11 endpoints are joined to oracle.local to generate domain authentication and endpoint

telemetry. Dedicated enterprise user accounts are used for later password spraying and privilege-boundary testing.

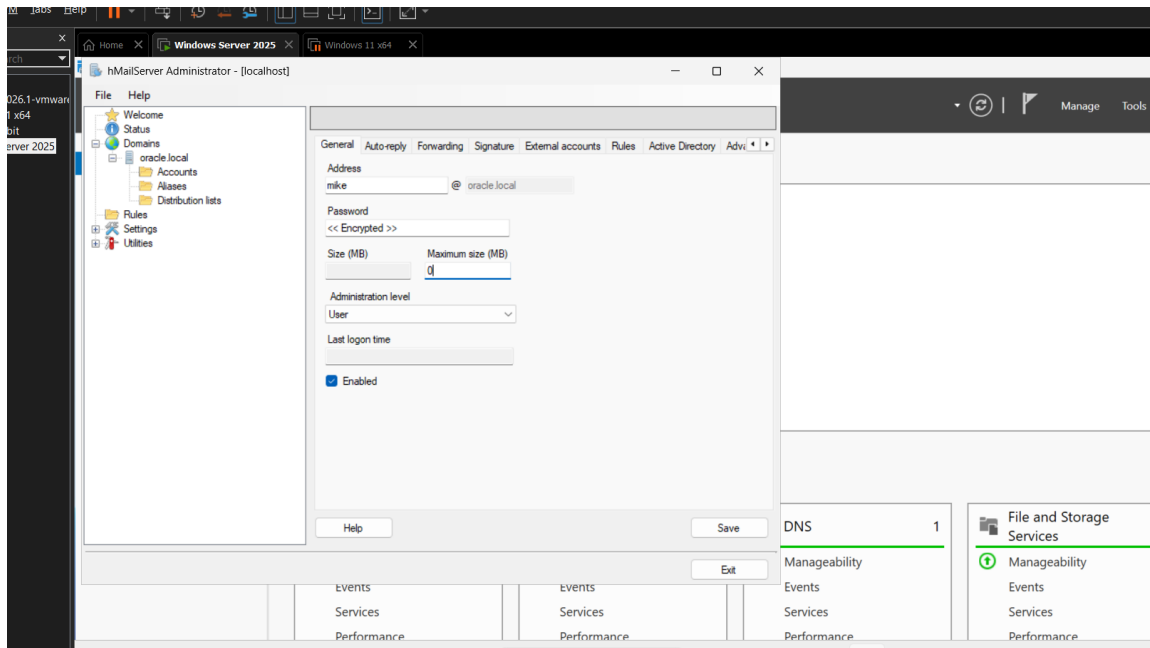


Figure 2. Active Directory user creation evidence.

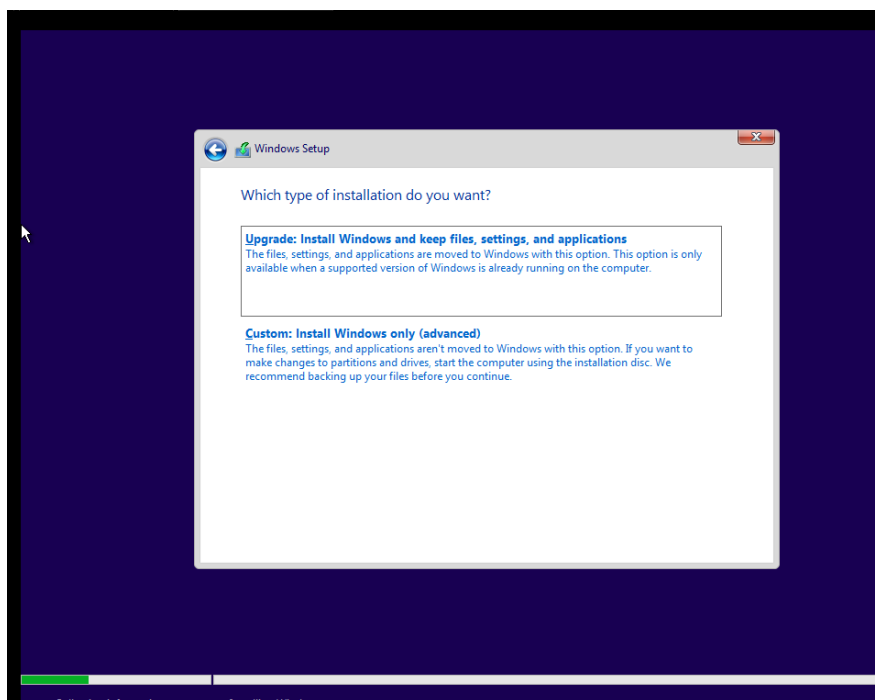


Figure 3. Windows endpoint aligned to oracle.local DNS suffix and internal network addressing.

3. Corporate Mail Infrastructure

hMailServer provides internal IMAP/SMTP mail flow. Thunderbird is used as the mail client for validation and later SOC alert-routing tests. The mail layer makes the lab more realistic because alert notifications, phishing simulations, and user-mail workflows can be tested without public infrastructure.

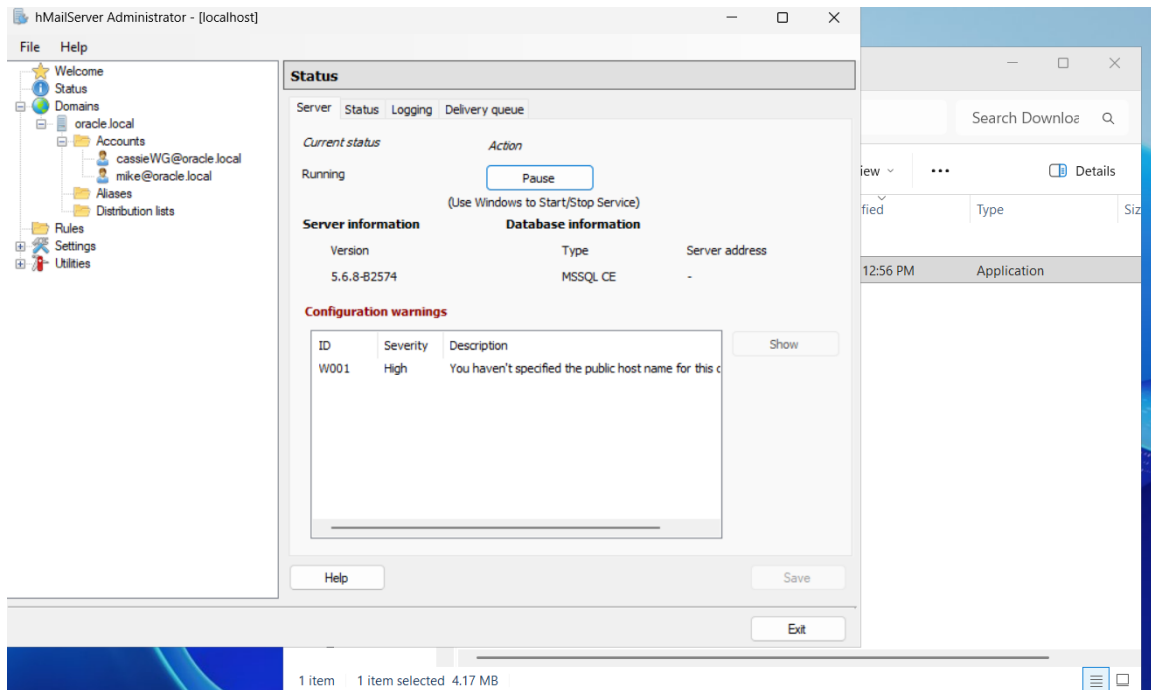


Figure 4. hMailServer service and mailbox status.

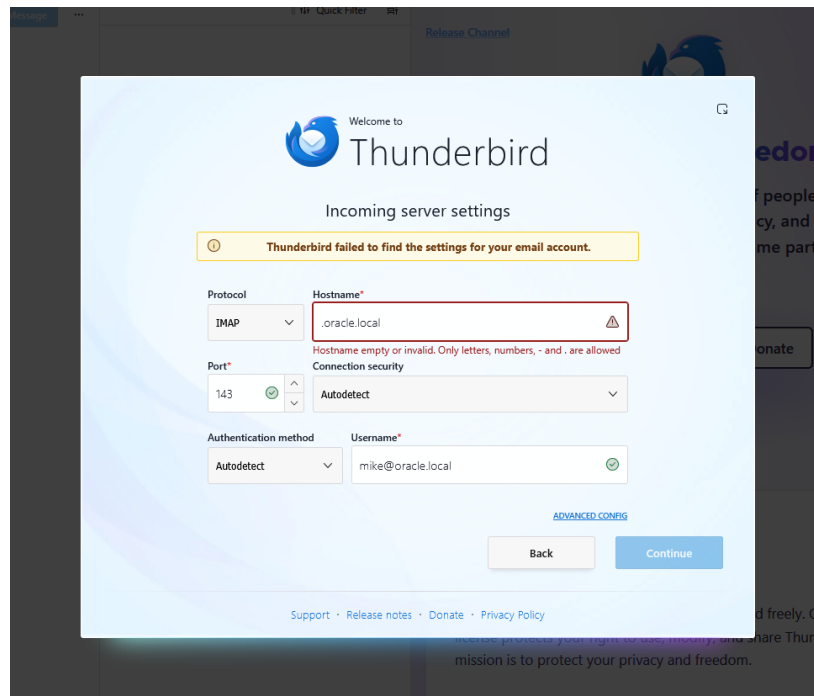


Figure 5. Thunderbird troubleshooting during IMAP discovery.

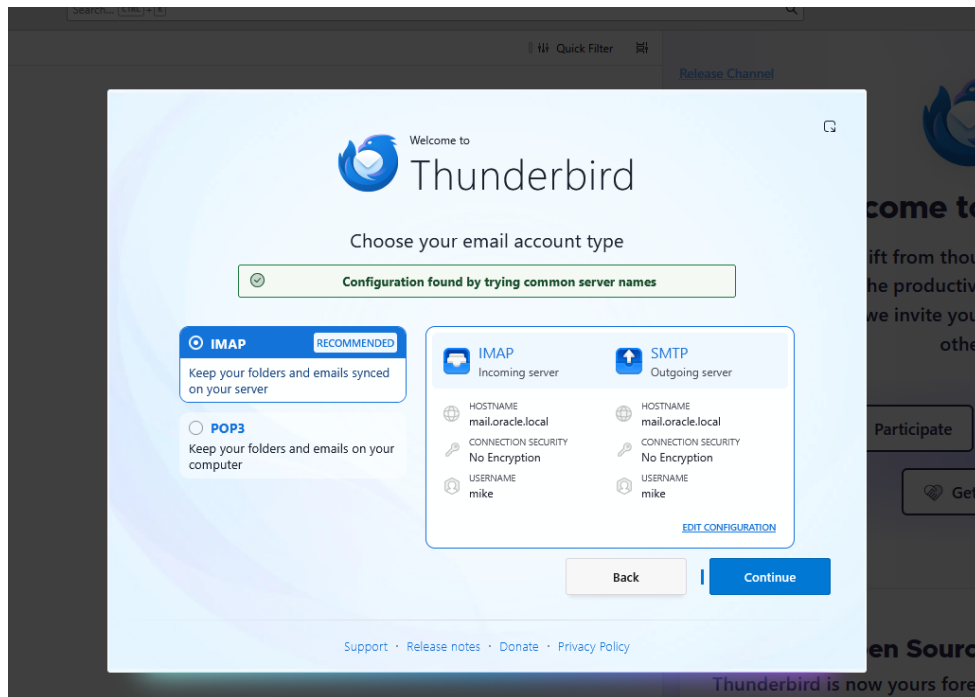


Figure 6. Thunderbird configuration validated after manual IMAP/SMTP parameters.

4. Linux Corporate Web and Domain Integration

The Debian corp-web endpoint extends the lab beyond Windows. It acts as a corporate intranet host and Linux telemetry source. DNS behavior was hardened by forcing the host toward the Active Directory DNS server and correcting the .local multicast DNS lookup order.

```
ping: oracle.local: Name or service not known
kara@corp-web:~$ sudo nmcli connection modify "Wired connection 1" ipv4.ignore-auto-dns yes
kara@corp-web:~$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
kara@corp-web:~$ cat /etc/resolv.conf
# Generated by NetworkManager
search oracle.local
nameserver 192.168.241.131
kara@corp-web:~$ ping -c 4 oracle.local
ping: oracle.local: Name or service not known
kara@corp-web:~$ ping oracle.local
ping: oracle.local: Name or service not known
kara@corp-web:~$
```

Figure 7. Linux .local DNS failure before remediation.

```
ping: oracle.local: Name or service not known
kara@corp-web:~$ ping oracle.local
ping: oracle.local: Name or service not known
kara@corp-web:~$ sudo nano /etc/nsswitch.conf
kara@corp-web:~$ ping -c 4 oracle.local
PING oracle.local (192.168.241.131) 56(84) bytes of data:
64 bytes from 192.168.241.131: icmp_seq=1 ttl=128 time=0.480 ms
64 bytes from 192.168.241.131: icmp_seq=2 ttl=128 time=0.442 ms
64 bytes from 192.168.241.131: icmp_seq=3 ttl=128 time=0.441 ms
64 bytes from 192.168.241.131: icmp_seq=4 ttl=128 time=0.385 ms

--- oracle.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 0.385/0.437/0.480/0.033 ms
kara@corp-web:~$ ^C
kara@corp-web:~$ ping oracle.local
PING oracle.local (192.168.241.131) 56(84) bytes of data:
64 bytes from 192.168.241.131: icmp_seq=1 ttl=128 time=0.321 ms
64 bytes from 192.168.241.131: icmp_seq=2 ttl=128 time=0.443 ms
64 bytes from 192.168.241.131: icmp_seq=3 ttl=128 time=0.444 ms
^C
--- oracle.local ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.321/0.402/0.444/0.057 ms
kara@corp-web:~$
```

Figure 8. Linux DNS resolution restored after configuration changes.

```
* Installing necessary packages: adcli
kara@corp-web:~$ sudo realm list
oracle.local
  type: kerberos
  realm-name: ORACLE.LOCAL
  domain-name: oracle.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@oracle.local
  login-policy: allow-realm-logins
kara@corp-web:~$
```

Figure 9. realmd discovery confirms Active Directory visibility from Debian.

5. Wazuh Agent Deployment on Operating Systems

The Wazuh agent layer is part of the completed lab build. It is already deployed. Agents are deployed across Windows and Linux operating systems so the lab can centralize endpoint telemetry in the Wazuh dashboard.

Agent / asset	Operating system	Telemetry focus
windows-server-dc / ORACLE-DC1	Windows Server 2025	AD authentication, Kerberos, account/group changes, Windows Event Logs
windows-10-vm / CASSIE-PC	Windows 10 Enterprise	Logon/logoff, endpoint activity, local security events
windows-11-vm / MIKE	Windows 11 Enterprise	Logon/logoff, endpoint activity, local security events
corp-web	Debian 13 XFCE	auth.log, Apache2, sudo, services, Linux system events
admin-node	Xubuntu Minimal	admin sessions, sudo, shell activity, privileged workstation telemetry

Deploy new agent

Select the package to download and install on your system:

 LINUX

☐ RPM amd64

☐ RPM aarch64

☐ DEB amd64

☐ DEB aarch64

 WINDOWS

☒ MSI 32/64 bits

 macOS

☐ Intel

☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

Figure 10. Wazuh dashboard agent deployment workflow for Windows MSI endpoints.

1 Select the package to download and install on your system:

LINUX

☐ RPM amd64
 ☐ RPM aarch64
 ☐ DEB amd64
 ☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel
 ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

2 Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Figure 11. Wazuh dashboard agent deployment workflow for Linux packages.

6. Wazuh Telemetry and Threat Hunting

Wazuh parses Windows Event Logs and Linux syslog/authentication activity into searchable events. The current baseline is suitable for tuning noisy Windows events, validating Linux authentication visibility, and building detection cases around authentication abuse and privilege changes.

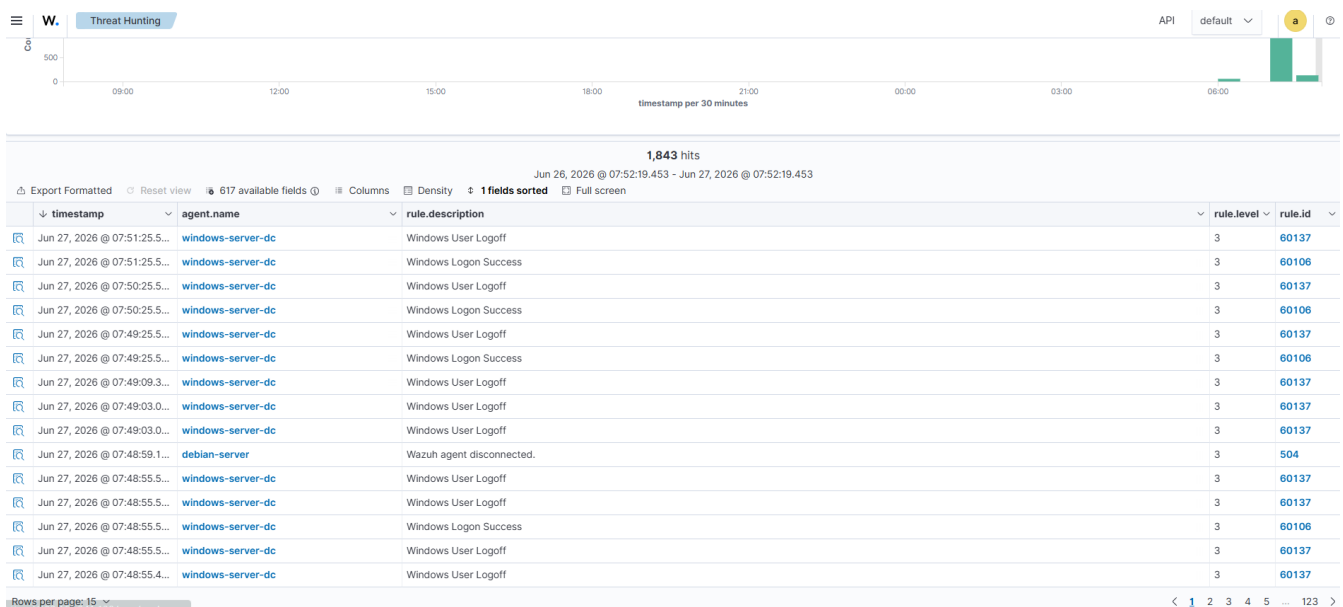


Figure 12. Wazuh Threat Hunting view showing parsed telemetry and agent names.

Alerting Monitors **Create monitor**

Create monitor

Monitor details

Monitor name

Monitor type

☐ Per query monitor
Per query monitors run a query and generate alerts based on trigger criteria that match query results.

☒ Per bucket monitor
Per bucket monitors run a query that evaluates trigger criteria based on aggregated values in the dataset.

☐ Per cluster metrics monitor
Per cluster metrics monitors run API requests to monitor the cluster's health.

☐ Per document monitor
Per document monitors run queries that return individual documents matching the trigger conditions.

☐ Composite monitor
Composite monitors chain the outputs of different monitor types and focus trigger conditions to reduce alert noise and generate finer results.

Monitor defining method

Specify the way you want to define your query and triggers. [Learn more](#)

☒ Visual editor
 ☐ Extraction query editor

7. Detection Validation Next

The build phase now moves from installation into validation. The next work focuses on controlled detection cases, not agent deployment.

- Password spraying against Active Directory users with Wazuh evidence and analyst notes.
- SSH brute-force against Linux endpoints with auth.log and Wazuh event proof.
- Privileged group membership modification in Active Directory.
- Local privilege-escalation indicators such as service creation, scheduled task creation, sudoers changes, and suspicious PowerShell.
- SMTP alert routing through hMailServer and Thunderbird for high-severity Wazuh alerts.

8. Repository Hygiene

The public repository intentionally excludes passwords, generated Wazuh credentials, root/user setup screenshots, private keys, certificates, installer tarballs, and raw logs containing sensitive values. Screenshots are included only when they match the documented phase and do not expose credentials.